

This script computes the median and maximum drought days experienced by each LAU and NUTS-3 between 2012 and 2026

It uses the coverage statistics computed in the previous step and combines them with the CDI dataset

```
import numpy as np
import re
import matplotlib.pyplot as plt
import geopandas as gpd
import pandas as pd
import rasterio
import glob
import os
import utils
```

```
from tqdm.notebook import tqdm
```

```
tqdm.pandas()
```

```
cdi_files = sorted(
    glob.glob(
        os.path.join(
            utils.raw_data_dir, "combined_drought_indicator_2012-2025", "**", "*.tif"
        )
    )
)
cdi_folders = sorted(
    glob.glob(
        os.path.join(utils.raw_data_dir, "combined_drought_indicator_2012-2025", "**")
    )
)
```

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)

# like before, toggle this variable to switch between LAU and NUTS-3 level of analysis
LAU_level = False
if LAU_level:
    coverage_files = glob.glob(
        os.path.join(
            utils.intermediate_data_dir, "cdi_raster_coverage_lau", "*.geojson"
        )
    )
    out_file = "drought_days_lau.geojson"
else:
    coverage_files = glob.glob(
        os.path.join(
            utils.intermediate_data_dir, "cdi_raster_coverage_nuts3", "*.geojson"
        )
    )
    out_file = "drought_days_nuts3.geojson"

# load all country-files into a single DataFrame
df = [gpd.read_file(f) for f in coverage_files]
lau_crs = df[0].crs

df = gpd.GeoDataFrame(pd.concat(df, ignore_index=True))
df = df.set_crs(lau_crs)

# load the first cdi file to extract metadata
with rasterio.open(cdi_files[0]) as src:
    geotiff = src.read(1)
    cdi_dims = src.read(1).shape

    assert src.crs == lau_crs

# compute pixel indices in the raster for each pair of x/y coordinates
min_coords = np.array(
    [src.index(x, y) for x, y in df[["xmin", "ymin"]].values.astype(float)]
)
max_coords = np.array(
    [src.index(x, y) for x, y in df[["xmax", "ymax"]].values.astype(float)]
)

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# enter pixel indices into the dataframe for later reference
df["row_start"] = min_coords[:, 0]
df["col_start"] = min_coords[:, 1]
df["row_end"] = max_coords[:, 0]
df["col_end"] = max_coords[:, 1]

# minor type cleanup
for col in ["col_start", "col_end", "row_start", "row_end"]:
    df[col] = df[col].astype(int)
df["cell_coverage"] = df["cell_coverage"].apply(lambda x: np.array(eval(x)))

def calculate_cell_weights(cell_coverage):
    # by default, exactextract returns the fraction of each grid cell that is being covered
    # We however desire the inverse: How much of the Polygon's area lies in a given grid cell.
    # The latter allows us to weigh grid values to determine the average value inside the LAU
    total_coverage = cell_coverage.sum(axis=None)
    return cell_coverage / total_coverage

df["cell_weights"] = df["cell_coverage"].apply(calculate_cell_weights)

# compute two 3D-matrices containing the total number of "warning" and "alert" days per pixel
warning_days = np.zeros(((2026 - 2012), cdi_dims[0], cdi_dims[1]))
alert_days = np.zeros(((2026 - 2012), cdi_dims[0], cdi_dims[1]))

for i, year in enumerate(range(2012, 2026)):
    print("processing year: ", year)

    year_folder = cdi_folders[i]
    assert bool(
        re.search(f"{year}01\d\d_{year}12", year_folder)
    ) # I messed this up once before so better be safe

    year_files = sorted(glob.glob(os.path.join(year_folder, "*.tif")))
    yearly_data = np.zeros(
        shape=(len(year_files), cdi_dims[0], cdi_dims[1]), dtype=np.uint8
    )

    for file_idx, path in enumerate(year_files):
        with rasterio.open(path) as src:
            yearly_data[file_idx] = src.read(1)

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# value 2 corresponds to CDI level "Warning", value 3 corresponds to level "Alert".
# See page 3 https://drought.emergency.copernicus.eu/data/factsheets/factsheet\_combinedD
cur_year_warning = yearly_data == 2
warning_days[i] = cur_year_warning.sum(axis=0) * 10

cur_year_alert = yearly_data == 3
alert_days[i] = cur_year_alert.sum(axis=0) * 10

```

```

processing year: 2012
processing year: 2013
processing year: 2014
processing year: 2015
processing year: 2016
processing year: 2017
processing year: 2018
processing year: 2019
processing year: 2020
processing year: 2021
processing year: 2022
processing year: 2023
processing year: 2024
processing year: 2025

```

```

def calculate_yearly_stats(row):
    """generates three sets of arrays for each row in the dataframe: "warning days", "alert o

    Args:
        row (pd.Series): a row in the dataframe, representing either a LAU or NUTS-3

    Returns:
        pd.Series: the same row, including three new columns
    """
    cell_weights = row["cell_weights"]
    width, height = cell_weights.shape

    # NOTE: @factcheck: I messed this up previously. I believe it to be right like this, but
    x0, x1 = row["row_start"] - width, row["row_start"]
    y0, y1 = row["col_start"], row["col_start"] + height

    # take the subset of the 3D-arrays that overlaps with the shape of this region
    # then, multiply each value by the amount of the total area it comprises

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# finally, summ over axes 2 and 3 to get a 1D-array representing the "warning" and "alert"
warning_days_per_year = (
    (warning_days[:, x0:x1, y0:y1] * cell_weights).sum(axis=(1, 2)).round(2)
)
alert_days_per_year = (
    (alert_days[:, x0:x1, y0:y1] * cell_weights).sum(axis=(1, 2)).round(2)
)

row["warning_days"] = warning_days_per_year.tolist()
row["alert_days"] = alert_days_per_year.tolist()
row["drought_days"] = (
    (warning_days_per_year + alert_days_per_year).round(2).tolist()
)
return row

print("calculating yearly stats...")
df = df.progress_apply(calculate_yearly_stats, axis=1)

```

calculating yearly stats...

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```

# compute median value, maximum value and the year that the maximum value occurred for each
for value in ["warning", "alert", "drought"]:
    df[f"median_{value}_days"] = df[f"{value}_days"].apply(
        lambda x: np.median(x).round(2)
    )
    df[f"max_{value}_days"] = df[f"{value}_days"].apply(lambda x: np.max(x).round(2))
    df[f"max_{value}_days_year"] = df[f"{value}_days"].apply(
        lambda x: 2012 + np.argmax(x)
    )

```

```

# drop columns that are no longer needed
if LAU_level:
    out_data = gpd.GeoDataFrame(
        df[
            [
                "CNTR_CODE",
                "LAU_NAME",
                "geometry",
            ]
        ]
    )

```

```

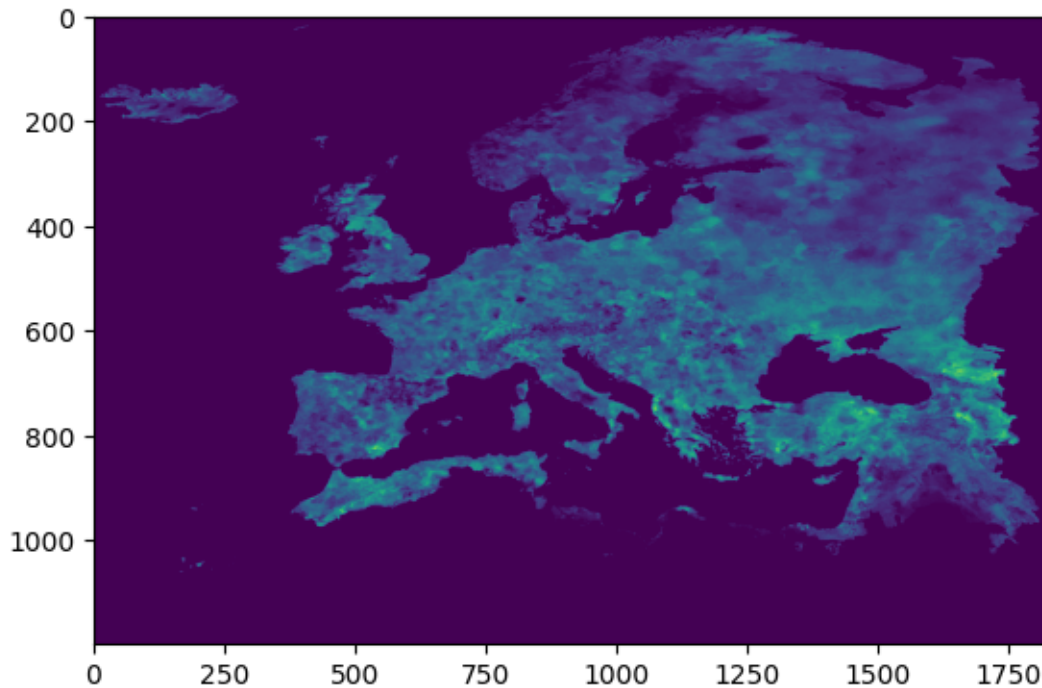
        "warning_days",
        "median_warning_days",
        "max_warning_days",
        "max_warning_days_year",
        "alert_days",
        "median_alert_days",
        "max_alert_days",
        "max_alert_days_year",
        "drought_days",
        "median_drought_days",
        "max_drought_days",
        "max_drought_days_year",
    ]
]
)
else:
    out_data = gpd.GeoDataFrame(
        df[
            [
                "NUTS_ID",
                "CNTR_CODE",
                "NAME_LATN",
                "NUTS_NAME",
                "geometry",
                "warning_days",
                "median_warning_days",
                "max_warning_days",
                "max_warning_days_year",
                "alert_days",
                "median_alert_days",
                "max_alert_days",
                "max_alert_days_year",
                "drought_days",
                "median_drought_days",
                "max_drought_days",
                "max_drought_days_year",
            ]
        ]
    )

out_data = out_data.to_crs(epsg=4326)
out_data.to_file(os.path.join(utils.out_data_dir, out_file))

```

Sanity check: plot the warning days as an indicator that we are producing sensible data

```
plt.imshow(warning_days.sum(axis=0))
```



Sanity check: Plot some weight matrices and their corresponding shapefiles to validate that the overlap computation makes sense

```
# --- Plot ---
fig, axs = plt.subplots(3, 3, figsize=(12, 12), squeeze=True)
axs = axs.flatten()

for i, ax in enumerate(axs):
    temp = df.iloc[np.random.randint(100) + (i * 100)]
    col_start = temp["col_start"]
    col_end = temp["col_end"]
    row_start = temp["row_start"]
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row_end = temp["row_end"]

cell_weights = temp["cell_weights"]

tif_local = geotiff[row_start:row_end, col_start:col_end]
# 1. Show raster using extent [x_min, x_max, y_min, y_max]
im = ax.imshow(
    cell_weights,
    extent=[temp["xmin"], temp["xmax"], temp["ymin"], temp["ymax"]],
    cmap="viridis",
    alpha=0.8,
)

gpd.GeoSeries(temp.geometry).plot(
    ax=ax, facecolor="none", edgecolor="red", linewidth=2
)

plt.colorbar(im, ax=ax, label="Cell weight")
if LAU_level:
    ax.set_title(temp["LAU_NAME"])
else:
    ax.set_title(temp["NUTS_NAME"])
plt.tight_layout()
plt.show()

```

